

DCP Challenge Ideas

DCP University Engineering Challenge — Category 2 Ideas

Predictive Maintenance & Reliability Early-Warning System

- Team:** Mubarak (Civil Eng — AI/ML, software, data analytics, web dev) + Wale (Mechanical Eng — hardware simulation, engineering design, AI/ML)
- Deadline:** Thursday, 9 July 2026 · 11:59 PM WAT (~7 days from now)
- Prize Pool:** ■1.5M (1st: ■700K · 2nd: ■500K · 3rd: ■300K)

What the Judges Want (Scoring Breakdown)

Criteria	Weight	What to Nail
Problem Diagnosis & Insight	30%	Deep current-state analysis, strong root-cause reasoning, smart use of data/assumptions
Solution Quality & Feasibility	30%	Sound engineering solution that actually works in a real cement plant
Business Case & Impact	20%	Clear value to DCP — cost savings, production gains, realistic roadmap
Risk, Safety & Sustainability	10%	Risk register, safety assessment, sustainability considerations

Communication &
Presentation

10%

Clean 10-slide pitch,
professional supporting
materials

Required Deliverables

1. 10-Slide Executive Presentation (strict structure — see below)

Technical Document containing:

- Technical appendix / model / prototype
- One-page business case
- Implementation roadmap (immediate → short → medium-term)
- Risk & safety assessment
- Statement of assumptions and data limitations

5 Track-Specific Deliverables:

1. Reliability early-warning framework
2. Risk-scoring model with clear assumptions
3. Dashboard mock-up or prototype
4. Example maintenance escalation workflow
5. Implementation roadmap for plant use

Top 5 Research Ideas (Ranked by Winning Potential)

IDEA 1: “CementGuard” — AI-Powered Multi-Sensor Fusion Early-Warning System

STRONGEST IDEA — Best fit for your combined skills

Concept

Build an AI/ML-based predictive maintenance system that fuses data from **vibration sensors, temperature probes, acoustic emission sensors, and motor current analysis** across DCP's critical equipment (kilns, mills, crushers, packers). The system uses anomaly detection + degradation trend models to produce a **Risk Priority Number (RPN)** for each equipment unit, displayed on a real-time dashboard.

Why This Wins

- Directly addresses the competition brief with a **practical, implementable** solution
- Combines both your skills perfectly: **Wale** handles equipment failure mode analysis (FMEA) + mechanical simulation, **Mubarak** builds the ML model + dashboard prototype
- Demonstrates **both engineering depth AND tech innovation** — exactly what stands out

Technical Approach

1. **Failure Mode Analysis (Wale):** Conduct FMEA on each equipment type (crusher, raw mill, coal mill, kiln, cement mill, packer). Identify the top 5 failure modes per equipment — bearing failure, shaft misalignment, thermal fatigue, refractory wear, gearbox degradation
2. **Sensor Architecture (Both):** Design a practical IoT sensor layout — vibration (accelerometers), temperature (thermocouples), acoustics (ultrasonic), motor current (CTs). Use publicly available industrial sensor specs
3. **ML Anomaly Detection Model (Mubarak):** Build a model using **Isolation Forest + LSTM autoencoder** on simulated/synthetic sensor data. Train on “normal” operating patterns, flag deviations as early warnings
4. **Risk Scoring Engine:** Composite score = $f(\text{vibration_anomaly}, \text{temp_trend}, \text{acoustic_signature}, \text{current_deviation}, \text{equipment_age}, \text{maintenance_history})$. Output a color-coded risk level (Green/Yellow/Orange/Red)
5. **Dashboard Prototype (Mubarak):** Build a working web dashboard showing real-time equipment health, risk scores, alert history, and recommended maintenance actions
6. **Maintenance Escalation Workflow (Both):** Define a clear SOP — when risk hits Yellow, auto-notify maintenance team; Orange triggers inspection within 48hrs; Red triggers immediate shutdown

Wale's Role

- FMEA analysis for all 6 equipment types
- Vibration analysis theory and threshold calculations
- Mechanical simulation of bearing degradation curves using MATLAB/ANSYS
- Equipment wear models and failure progression timelines
- Sensor placement optimization based on mechanical knowledge

Mubarak's Role

- ML model architecture and training pipeline
- Synthetic data generation mimicking real sensor outputs
- Web-based dashboard prototype (React/Next.js or pure HTML/JS)
- Risk scoring algorithm implementation
- Data visualization and alert system

Key Data Sources (No Real DCP Data Needed!)

- NASA C-MAPSS turbofan degradation dataset (proxy for rotating equipment)
- Bearing vibration datasets from Case Western Reserve University
- Published cement plant KPIs from industry reports
- Dangote Cement annual reports for production capacity data

IDEA 2: “DigitalTwin-Predict” — Digital Twin-Based Kiln Health Monitoring

Concept

Create a **digital twin** of a rotary cement kiln that simulates thermal stress, refractory wear, and shell deformation in real-time. The twin compares expected vs. actual performance to predict failures 2-4 weeks in advance.

Why It's Strong

- **Kiln** is the most expensive and critical equipment in any cement plant — a kiln failure costs \$50K-\$200K+ per incident
- Digital twin concept is cutting-edge and impressive to judges
- Plays perfectly to Wale's simulation skills

Technical Approach

1. Model kiln thermodynamics (heat transfer, refractory degradation)
2. Simulate shell temperature distribution using FEM (Wale)
3. Compare simulated vs. "measured" temperatures to detect hot spots
4. ML regression model to predict remaining useful life (RUL) of refractory lining
5. Dashboard showing 3D kiln visualization with heat map overlays

Risk

- More complex, potentially harder to prototype convincingly in 7 days
- Narrower scope (kiln only vs. all equipment)

IDEA 3: "ReliabilityPulse" — Failure Pattern Recognition from Maintenance Logs

Concept

Design an NLP + data analytics system that mines **maintenance work order logs** (text-based) to identify recurring failure patterns, seasonal trends, and hidden correlations between equipment failures. Uses text classification + time-series analysis.

Why It's Strong

- Many cement plants already have years of maintenance log data
- Low-cost implementation (no new sensors required)

- Shows data-driven thinking that judges love

Technical Approach

1. Simulate realistic maintenance log dataset (text entries)
2. NLP pipeline: text preprocessing → named entity recognition (equipment, failure type) → classification
3. Pattern mining: association rules to find “if X fails, Y fails within 14 days”
4. Trend analysis: seasonal decomposition of failure frequency
5. Dashboard with failure prediction calendar and maintenance scheduling optimizer

Risk

- Less “engineering” and more “data science” — may not impress engineering judges as much
- Harder to demonstrate mechanical depth

IDEA 4: “VibrationIQ” — Low-Cost Vibration-Based Condition Monitoring

Concept

Design a **cost-effective** vibration monitoring system using MEMS accelerometers (like the ADXL345) connected to ESP32 microcontrollers. Focus on detecting bearing defects, imbalance, and misalignment in mills and crushers through FFT frequency analysis.

Why It’s Strong

- Extremely practical and affordable (sensors < \$10 each)
- Directly applicable to Nigerian cement plants with budget constraints
- Hardware + software integration shows engineering breadth

Technical Approach

1. Design sensor node hardware (ESP32 + accelerometer + WiFi)
2. FFT analysis to detect characteristic bearing defect frequencies (BPFO, BPFI, BSF, FTF)
3. ISO 10816 vibration severity classification
4. Edge computing: anomaly detection on the microcontroller itself
5. Central dashboard aggregating data from all sensor nodes

Wale's Role: Vibration theory, bearing defect frequency calculations, mechanical mounting design

Mubarak's Role: Embedded firmware, FFT algorithm, cloud dashboard

Risk

- More narrow in scope
- Hardware prototype may be hard to demonstrate without physical equipment

IDEA 5: “ClinkerSafe” — Thermal Imaging + AI for Kiln Shell Monitoring

Concept

Use **infrared thermal imaging** (or simulated thermal data) + computer vision AI to automatically detect hot spots on kiln shells, which indicate refractory wear. The system captures periodic thermal scans and uses image classification to rate refractory health.

Why It's Strong

- Visual and impressive for presentations
- Hot spot detection is a real, validated technique in cement plants
- Computer vision is a strong AI application

Technical Approach

1. Simulate thermal images of kiln shells (healthy vs. degraded)

- 2. CNN image classifier to detect and categorize hot spots
- 3. Severity scoring based on hot spot size, temperature, and growth rate
- 4. Alert system with maintenance recommendations
- 5. Historical tracking to predict refractory replacement schedule

My Recommendation: GO WITH IDEA 1 (“CementGuard”)

Here’s why:

Factor	Idea 1 (CementGuard)	Others
Covers ALL 5 required deliverables	Yes	Partially
Uses both your skills fully	Yes (Mubarak: ML + Dashboard, Wale: FMEA + Simulation)	Varies
Feasible in 7 days	Yes (With focused execution)	Some are tight
Impressive to judges	Yes (Working prototype + engineering rigor)	Varies
Addresses full scope (all equipment)	Yes (Crushers, mills, kilns, packers)	Some are narrow
Business case is clear	Yes (Cost of unplanned downtime is huge)	Yes
Differentiator	AI + Engineering hybrid approach	Less unique

Suggested 7-Day Execution Plan for Idea 1

Day	Mubarak	Wale
Day 1 (Wed)	Research ML models for predictive maintenance, collect proxy datasets	FMEA analysis on all 6 equipment types, failure mode research
Day 2 (Thu)	Design risk scoring algorithm, start synthetic data generation	Vibration analysis theory, bearing defect frequency calcs, sensor specs
Day 3 (Fri)	Build ML anomaly detection model (Isolation Forest + LSTM)	Mechanical simulation of degradation curves, sensor placement design
Day 4 (Sat)	Build dashboard prototype (front-end + visualization)	Design maintenance escalation workflow, write technical feasibility section
Day 5 (Sun)	Integrate ML model with dashboard, test alert system	Write business case, cost-benefit analysis, implementation roadmap
Day 6 (Mon)	Polish dashboard, create demo data, record demo video	Build 10-slide presentation, risk & safety assessment
Day 7 (Tue)	Final review, fix bugs, compile technical document	Final review, proofread, format, SUBMIT before 11:59 PM

Pro Tips to Win

- Use real industry numbers:** Dangote Cement's annual report shows production capacity, number of plants, etc. Use these for business case calculations
- Show a WORKING prototype:** Even a basic dashboard with simulated data running live will massively impress judges over static mockups
- Name your solution:** "CementGuard" or similar makes it memorable and professional

4. **Quantify everything:** “This system can reduce unplanned downtime by 30-40%, saving ■X million per year per plant”
5. **Address Nigeria-specific challenges:** Power instability, internet connectivity, skill gaps in maintenance teams — show you’ve thought about real deployment
6. **Practice the pitch:** 10 slides, rehearse timing, anticipate judge questions

[!IMPORTANT] The deadline is **9 July 2026 at 11:59 PM WAT** — you have roughly 7 days. Start immediately!